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The MedRank algorithm and Spearman's footrule versus the Borda count and Spearman's rank correlation

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ABSTRACT

Social welfare functions can be defined or obtained as solutions of distance minimization problems. There are well-known restricted connections between the MedRank algorithm and Spearman's footrule, and the Borda count and Spearman's rank correlation, respectively. We show that these connections are of different strengths since the former is valid only on a small set of profiles, while the latter is valid on all profiles.

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1. Introduction

In this paper we follow the 'operations research approach', which strives for selecting voting rules, also related to the more general social welfare functions (SWF), as solutions to appropriately defined distance minimization problems. In fact, the Kemény–Young method [1] is defined as the SWF minimizing the Kendall τ distance, the Borda count can be derived as the one minimizing the Spearman rank correlation distance and the MedRank rule can be derived as the one minimizing the Spearman footrule [2]. The limitations of the latter two results are also contained in Dwork et al. [2]. In particular, they proved that the Borda count is obtained for any preference profile for which the Borda count gives a linear ordering, while the MedRank algorithm is obtained if the median ranks of the alternatives form a permutation. In a recent publication in Hungarian Bednay et al. [3] pointed out that these two results differ in their nature. In particular, as the number of voters increases the probability of ties in the case of the Borda count can be made arbitrarily small, while ties in the median ranks occur with probability arbitrarily close to one. In this paper, in our simulations we highlight the gap between the MedRank algorithm and the Spearman's footrule welfare function and we prove an even stronger result concerning the Borda count since it equals the Spearman's rank correlation welfare function for any number of voters. Clearly, an analogous statement immediately follows for the Kemény–Young method since it is defined as a solution of a distance minimization problem.

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The mentioned distance functions are well-known in mathematics and statistics. Diaconis and Graham [4] considered them as distances on permutations and in statistics they are used as measures of association for ordinal data (for more on historical notes we refer to [5]). Berry and Mielke [6] considered the extensions of the Spearman footrule and the Spearman rank correlation distances to rankings with tied ranks. For the Spearman rank correlation the associated optimal voting rule is the Borda count, while with respect to the Spearman footrule the MedRank algorithm is usually mentioned as the optimal one. However, as it will be clear from our results the latter one is only true on a small set of possible combinations of rankings. We also consider the Bucklin rule as the possible refinement of the MedRank algorithm.

The structure of the paper is as follows. Section 2 provides a literature overview. Section 3 introduces the basic notations and the employed metrics on the set of SWF. Section 4 presents the SWFs under study. Section 5 explains the computational scheme, presents and discusses the results obtained by simulations. Sections 6 and 7 deal with the MedRank algorithm and the Borda count, respectively. Finally, Section 8 concludes. Additional figures are relegated to Appendix 1, while the proof of Proposition 6.1 can be found in Appendix 2. For the sake of completeness Appendix 3 contains an earlier result by Bednay et al. [3] published only in Hungarian.

2. Literature overview

There is quite extensive literature on deriving voting rules as solutions of optimization problems related to desired properties. Farkas and Nitzan [7] derived the Borda count as the solution of an optimization problem on the set of social choice functions (SCFs), which selects only a winner and not a whole social ranking, by minimizing the distance from the unanimity principle. Taking other metrics, Nitzan [8] obtained the plurality rule among other rules. However, the obtained voting rules are functions of the respective distance functions and principle(s). Lehrer and Nitzan [9] and Campbell and Nitzan [10] showed that basically any voting rule can be distance rationalized. The approach of minimizing the distance from a set of profiles with a clear winner such as the unanimous winner, the majority winner, or the Condorcet winner has been developed further by Elkind et al. [11], Andjiga et al. [12] and Mahajne et al. [13] among others. Bednay et al. [14] offered a ‘dual’ approach based on distance maximization from the closest dictator in which the dictatorial rules are taken as the benchmark and the distances of SCFs from them are determined. When looking at voting rules being farthest away from the closest dictatorial rule, very undesirable voting rules were obtained, which would lead intuitively to the largest level of dissimilarity. Continuing this approach, Bednay et al. [15] introduced a non-dictatorship index and compared SCFs based on it.

From a historical perspective, a voting procedure based on median ranks named after Bucklin (as described in our paper) was first used in Grand Junction, Colorado, from 1909 to 1922. The procedure was used in a total of 55 cities in the United States. It was also used to elect the directors of the Federal Reserve. The procedure was proposed by Condorcet as early as 1793 and was used in Genoa in the eighteenth century. For historical details, see for example [16, p. 65]. It is an elementary relation discussed in the teaching of mathematics and statistics that the median minimizes the mean absolute deviation, so the result for the MedRank algorithm described in the first paragraph of Dwork et al. [2] is not surprising.

It is worth mentioning, that in the field of chemistry [17] introduces a ranking procedure using the sum of rank differences (SRD) to compare or combine different methods, which he finds more advantageous than other statistical procedures. His procedure is related to the Spearman's footrule distance. Sziklai and Héberger [18] use the SRD method to compare house apportionment methods and to address the political districting problem. Lin [19] discusses rank sum methods motivated primarily by biological and Internet applications. Erdélyi et al. [20] investigate the computational complexity of manipulating the Bucklin method.

The other rule we investigate in this paper is the Borda count, which gives the worst alternative 0 point, the second worst alternative 1 point, and so on. Winners are the alternatives with the highest overall score. This method can be traced back until Ramon Llull in 1283 and Nicolas de Cusa in 1433, but named after Jean-Charles de Borda, who proposed this method to the French Academy of Sciences in 1770 for electing its officers (see, for instance, Pukelsheim, [21]). The Borda count or related methods are used widely, in a few governmental elections, in educational institutions, professional organizations, in granting sport awards, etc. [16, p. 66].

3. The framework

Let $A = \{a_1, \dots, a_m\}$ be the set of alternatives and $N = \{1, \dots, n\}$ be the set of voters. We shall denote by $\mathcal{P}$ the set of all linear (or preference) orderings (irreflexive, transitive and total binary relations) on A . If $i \in N$, then $\succ_i \in \mathcal{P}$ is the preference ordering of voter i . We call $\Pi = (\succ_1, \dots, \succ_n) \in \mathcal{P}^n$ a preference profile, where $\mathcal{P}^n$ stands for the set of all preference profiles. We write $\succ_{-i}$ if we drop voter i 's preference ordering from profile Π . Then $\Pi = (\succ_i, \succ_{-i})$. Moreover, let $rk[a, \succ]$ denote the *rank* of alternative a in the ordering $\succ \in \mathcal{P}$ (i.e. $rk[a, \succ] = 1$ if a is the top alternative in the ranking $\succ$, $rk[a, \succ] = 2$ if a is second-best, and so on).

Definition 3.1: A mapping $f : \mathcal{P}^n \rightarrow \mathcal{P}$ that selects the (social) linear ordering is called a *social welfare function*, henceforth, SWF.

As our definition of an SWF does not allow for possible ties, while the formulas defining the well-known voting rules just determine weak orderings, we employ fixed anonymous¹ tie-breaking rules to break ties. A tie-breaking rule $\tau : \mathcal{P}^n \rightarrow \mathcal{P}$ maps preference profiles to linear orderings on A , which will be only employed when a formula leaves alternatives indifferent. If there are more tied alternatives at a certain position based on a formula 'almost' specifying an SWF, then the tied alternatives follow the ordering given by the tie-breaking rule. When determining the average distance for the SWFs under study, we employ the fixed (also called the lexicographic) tie-breaking rule $a_1 \tau(\Pi) a_2 \tau(\Pi) \dots \tau(\Pi) a_m$ for each profile $\Pi \in \mathcal{P}^n$, which is anonymous and does not depend on the actual preference profile $\Pi \in \mathcal{P}^n$. Henceforth, we will briefly write $a_1 \tau a_2 \tau \dots \tau a_m$ for the ordering of the alternatives by this tie-breaking rule.

We shall denote by $d : \mathcal{P} \times \mathcal{P} \rightarrow \mathbb{R}_+$ a quasidistance function or quasimetric above the set of preferences, which satisfies (i) $d(\succ, \succ') = 0$ if and only if $\succ = \succ'$, (ii) $d(\succ, \succ') = d(\succ', \succ)$ for any $\succ, \succ' \in \mathcal{P}$, and (iii) $d(\succ, \succ') \leq c(d(\succ, \succ'') + d(\succ'', \succ'))$ for any $\succ, \succ', \succ'' \in \mathcal{P}$ for a given $c > 0$ (called as a c -relaxed triangle inequality). We employ the following three

distance functions

$$d_1(\succ, \succ') = \sum_{i=1}^m |rk[a_i, \succ] - rk[a_i, \succ']|, \quad (1)$$

$$d_2(\succ, \succ') = \sum_{i=1}^m (rk[a_i, \succ] - rk[a_i, \succ'])^2, \quad (2)$$

$$d_B(\succ, \succ') = \#\{(a_i, a_j) \in A^2 \mid a_i \succ a_j \text{ and } a_j \succ' a_i\}, \quad (3)$$

where d_1 and d_B are a metrics ($c = 1$), while d_2 is just a quasimetric with $c = 2$.

It will be helpful for us that [4] determined for the Spearman footrule, the Spearman rank correlation and the Kendall τ (or Bubble sort) distance the maximum distances between two linear orderings, which equal

$$C_1 = \lfloor m^2/2 \rfloor, \quad C_2 = (m^3 - m)/3 \text{ and } C_B = (m - 1)m/2.$$

We extend d to distances between profiles and preferences in an additive way:

$$d_k(\Pi, \succ) = \frac{1}{n} \sum_{i=1}^n d_k(\succ_i, \succ),$$

where $\Pi = (\succ_1, \dots, \succ_n)$ and $k \in \{1, 2, B\}$. We define the average distance of a given SWF F by

$$D_k(F) = \sum_{\Pi \in \mathcal{P}^n} \frac{1}{C_k(m!)^n} d_k(\Pi, F(\Pi)).$$

4. Social welfare functions

We consider the following three common social welfare functions. To introduce these SWFs let $m = 5$, $n = 7$, $A = \{a_1, a_2, a_3, a_4, a_5\}$, $N = \{1, 2, \dots, 7\}$ and we pick the profile in Table 1.

Since there can be tied alternatives based on the formula partially defining an SWF we use the lexicographic tie-breaking rule to resolve ties in order to arrive to the linear ordering chosen by the respective SWF.

1. The *Borda count*, briefly denoted by BC , orders the alternatives based on the sum of their ranks. Let $sc[a, \succ] = m - rk[a, \succ]$ be the *Borda score* of an alternative a received by the ordering $\succ \in \mathcal{P}$. In particular, an alternative with a higher sum of Borda scores is

Table 1. A profile with 5 alternatives and 7 voters.

Rank	$\succ_1$	$\succ_2$	$\succ_3$	$\succ_4$	$\succ_5$	$\succ_6$	$\succ_7$
1	a_5	a_4	a_1	a_5	a_4	a_2	a_2
2	a_1	a_1	a_4	a_3	a_1	a_5	a_5
3	a_4	a_3	a_2	a_4	a_5	a_4	a_4
4	a_3	a_5	a_3	a_2	a_3	a_3	a_1
5	a_2	a_2	a_5	a_1	a_2	a_1	a_3

preferred over an alternative with a lower sum of Borda scores. In case of the profile in Table 1 the sum of the Borda scores of alternatives a_1, a_2, a_3, a_4 and a_5 are 14, 11, 9, 19 and 17, respectively. Therefore, the social ordering determined by BC is $a_4 \succ a_5 \succ a_1 \succ a_2 \succ a_3$. The SWF BC_τ is the *Borda count* if for all $\Pi = (\succ_i)_{i=1}^n \in \mathcal{P}^n$ and all pairs of distinct alternatives a and b we have

$$a BC_\tau (\Pi) b \Leftrightarrow \sum_{i=1}^n sc[a, \succ_i] > \sum_{i=1}^n sc[b, \succ_i] \text{ or } \left(\sum_{i=1}^n sc[a, \succ_i] = \sum_{i=1}^n sc[b, \succ_i] \text{ and } a \tau b \right).$$

2. The *Bucklin rule* determines for each alternative $a \in A$ the smallest h_a such that considering the first h_a ranked alternatives it appears more than $n/2$ times in a given profile. Looking at Table 1, we see that no alternative receives a majority (i.e. 4 votes) when counting only the numbers of top ranked alternatives. Now taking also the second ranked alternatives into consideration we see that both a_1 and a_5 appear four times, where the tie-breaking rule τ gives priority to a_1 . Thus, $h_{a_1} = h_{a_5}$. Admitting also third ranked alternatives, a_4 appears 7 times in the first three rows and we have $h_{a_4} = 3$. If we also take the fourth ranked alternatives into account a_3 and a_2 appear 6 and 4 times, respectively. Hence, $h_{a_2} = h_{a_3}$. In case of tied values h_{a_2} and h_{a_3} priority is given to the alternative with the higher number of occurrences, and if even these ones are the same, the tie-breaking rule τ is applied. Then the Bucklin rule gives the ranking $a_1 \succ a_5 \succ a_4 \succ a_3 \succ a_2$. For each alternative a the rank h_a is determined by the median voter in the increasingly reordered sequence of voters by ranks based on how they rank alternative a . We shall denote the Bucklin rule by BR . Then the SWF BR_τ is the *Bucklin rule* if for all $\Pi \in \mathcal{P}^n$ and all pairs of distinct alternatives we have

$$a BR_\tau (\Pi) b \Leftrightarrow \begin{aligned} &h_a < h_b \text{ or} \\ &(h_a = h_b \text{ and } \# \{i \in N \mid rk[a, \succ_i] \leq h_a\} > \# \{i \in N \mid rk[b, \succ_i] \leq h_b\}) \text{ or} \\ &(h_a = h_b \text{ and } \# \{i \in N \mid rk[a, \succ_i] \leq h_a\} = \# \{i \in N \mid rk[b, \succ_i] \leq h_b\} \\ &\text{and } a \tau b). \end{aligned}$$

3. The Bucklin rule has itself several variants, but there is a related one, which is called the *MedRank rule* in the computer science literature as an abbreviation for the aggregation rule based on the median ranks of the alternatives, henceforth also briefly *MR*. The modification lies in the fact that only the median ranks of the alternatives matter, that is only the above defined $h_{a_1}, \dots, h_{a_n}$ matter (see for instance, [2]). In our framework of SWFs for two distinct alternatives a and b for which $h_a = h_b$ the tie-breaking rules have to be employed immediately without taking into account how many times these two alternatives appear in the top h_a positions of all voters. Therefore, the MedRank rule is less decisive than the Bucklin rule and in fact has tied alternatives far more frequently. For the preference profile in Table 1, the MedRank rule determines the ranking $a_1 \succ a_5 \succ a_4 \succ a_2 \succ a_3$ with the tie-breaking rule τ employed in this section, which is different from the result of the Bucklin procedure. Dwork et al. [2] showed that at least for those profiles in which all median ranks of the alternatives are pairwise distinct (that is, they form a permutation),

Table 2. Footrule gives $a_5 \succ a_1 \succ a_4 \succ a_3 \succ a_2$ as a minimum cost matching.

	1	2	3	4	5
a_1	14	9	10	11	14
a_2	17	14	11	10	11
a_3	19	12	7	4	9
a_4	9	6	5	12	19
a_5	11	8	9	12	17

Table 3. Rank correlation gives $a_4 \succ a_5 \succ a_1 \succ a_2 \succ a_3$ as a minimum cost matching.

	1	2	3	4	5
a_1	44	23	16	23	44
a_2	61	34	21	22	37
a_3	57	26	9	6	17
a_4	17	6	9	26	57
a_5	31	16	15	28	55

the ranking obtained by the MedRank rule equals the ranking obtained by the Spearman footrule welfare function defined in the next point.

4. For a given profile $\Pi \in \mathcal{P}^n$ the *Spearman footrule welfare function* selects a ranking $\succ \in \mathcal{P}$ that minimizes $d_1(\Pi, \succ)$. The ranking obtained by the Spearman footrule welfare function can be found in polynomial time by solving a minimum cost perfect matching problem in a bipartite graph as already noted in Dwork et al. [2]. It can happen that the minimum cost perfect matching is not unique. Then a tie-breaking rule τ can help us to choose one matching from the optimal ones, for instance in a lexicographic way, that is select those matchings in which the first alternative is ranked higher by τ than the others, if not decisive, then from those matchings in which the first alternatives are ranked highest are the same select those matchings by τ in which the second alternative is ranked higher by τ , and so on. The solution for the profile in Table 1 is illustrated in Table 2. The entries in Table 2 specify for each alternative a_i and for each possibly assigned rank j the sum of the absolute value of rank differences, i.e. $\sum_{l=1}^n |rk[a_i, \succ_l] - j|$. The numbers in boldface highlight the minimum cost perfect matching of the profile in Table 1.

5. For a given profile $\Pi \in \mathcal{P}^n$ the *Spearman rank correlation welfare function* selects a ranking $\succ \in \mathcal{P}$ that minimizes $d_2(\Pi, \succ)$, which can also be found in polynomial time by solving a minimum cost perfect matching problem in a bipartite graph. It can happen that the minimum cost perfect matching is not unique. Then a tie-breaking rule τ can help us to choose one matching, for instance, in a lexicographic way, that is select the matching in which the first alternative is ranked higher by τ , if not decisive, then among the matchings in which the first alternatives are ranked highest are the same select by τ the matchings in which the second alternative is ranked higher by τ , and so on. The solution for the profile in Table 1 is illustrated in Table 3. The entries in Table 3 specify for each alternative a_i and for each possibly assigned rank j the sum of the squared rank differences, i.e. $\sum_{l=1}^n (rk[a_i, \succ_l] - j)^2$. The numbers in boldface highlight the minimum cost perfect matching of the profile in Table 1. We can see that we obtained the same preference relation as in case of the Borda count.

5. Simulations

The simulations in this section extend the ones in Bednay et al. [3] published in Hungarian by including the expected average distances for the Spearman footrule welfare function and by comparing the Borda count, the Bucklin rule and the MedRank rule based on the Spearman rank correlation and Kendall τ distances. While providing also exact values for a smaller number of voters, in our simulations for a larger number of voters we performed the estimation of the average distances for 3, 4 and 5 alternatives and up to 100 voters. We have generated 2500 random preference profiles, where each profile is chosen with equal probability. In this section we provide the graphs for 5 alternatives to illustrate the case, while the graphs for 3 and 4 alternatives can be found in Appendix 1 in case of the Spearman footrule.² Qualitatively, there are no relevant differences in case of 3 and 4 alternatives. We have coded our simulations and the distance functions in Python, which are available from the authors upon request. For excellent packages written in R and Python about modelling rankings we refer to Lee and Yu [22].

For three alternatives and a smaller number of voters (up to 7) we provide the exact values of the average distances for the Spearman footrule welfare function in Table 4. The exact values in the table are close to the results obtained by our simulations for three alternatives. We can observe that for at least 4 voters the MedRank rule is outperformed by the other two voting rules since smaller D_1 values are better.

Because of the slow rate of convergence in Figure 1 and the steepness of the three graphs in case of a small number of voters we show the graphs only ranging from 3 to 50 voters.³ Figure 1 underlines that the MedRank rule is outperformed by the other two voting rules. However, it is even more surprising that the Borda count outperforms the Bucklin rule, which is a kind of refinement of the MedRank rule.

This is in contrast to the result that the MedRank rule minimizes the Spearman footrule distance for those profiles for which the median ranks of the alternatives form a permutation of the ranks (see [2]). It is easy to see that a large set of such profiles does not satisfy this property. For instance, rank 1 as the median rank of an alternative emerges in a profile if and only if it has a strict majority of the top positions. m alternatives satisfy this property with probability

$$m \sum_{k=\lfloor n/2 \rfloor + 1}^n \binom{n}{k} \left(\frac{1}{m}\right)^k \left(\frac{m-1}{m}\right)^{n-k}.$$

For example, in case of $m = 3$ and $n = 10$ the above probability approximately equals 0.229691. Many profiles fail this criteria and there can be even other missing ranks, and furthermore this may not only be the case for a fixed alternative. The numerical example also shows that even with few voters, taking only the median rankings of the alternatives is unlikely to yield a linear ordering.

Table 4. Exact values of D_1 for $m = 3$.

SWF/ n	3	4	5	6	7
MedRank	0.370370	0.449074	0.464506	0.496914	0.513489
Borda	0.384259	0.439815	0.460648	0.482017	0.495842
Bucklin	0.370370	0.449074	0.453704	0.494663	0.493377
Footrule	0.370370	0.414352	0.444444	0.464763	0.479124

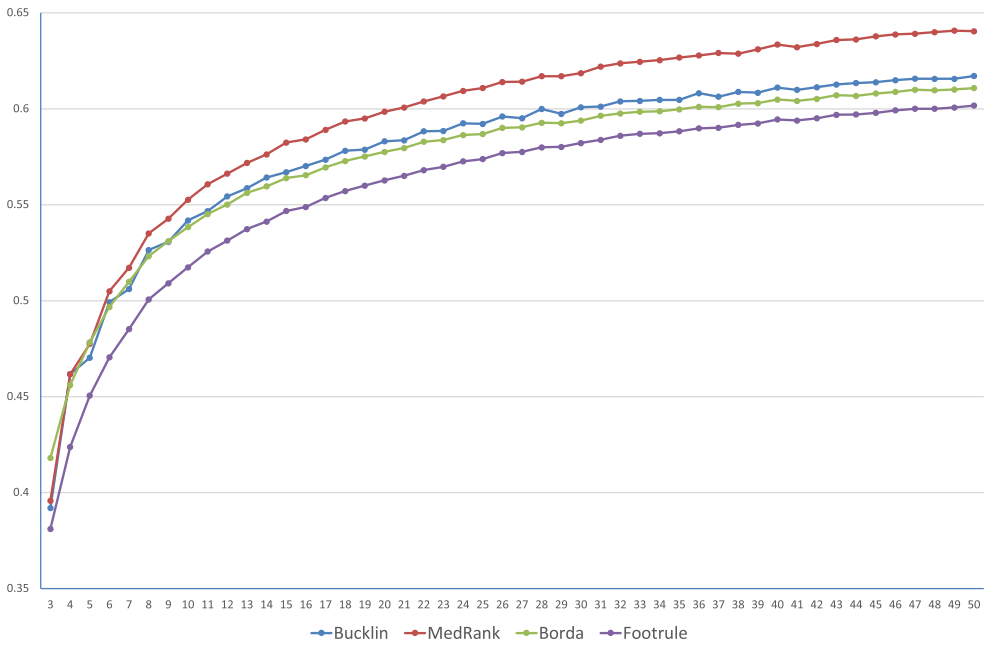


Figure 1. Average distance in case of D_1 and $m = 5$.

Table 5. Exact values of D_2 for $m = 3$.

SWF \ n	3	4	5	6	7
MedRank	0.263889	0.310764	0.334491	0.354681	0.372471
Bucklin	0.256944	0.305556	0.318287	0.342303	0.349055
Borda	0.25	0.288194	0.309414	0.326710	0.339463
Rank correlation	0.25	0.288194	0.309414	0.326710	0.339463

In order to check how the Borda count, the Bucklin rule and the MedRank rule perform with respect to the Spearman rank correlation distance we provide the exact values of the average distances for three alternatives in Table 5. Bednay et al. [3] showed that for a large number of voters the expected average distances of the Spearman rank correlation welfare function and the Borda count have to be close to each other. However, Table 5 indicates that they are already identical in the case of a small number of voters, which lead to conjecturing Proposition 7.1. In addition, we can observe in Table 5 that with respect to the expected average distance the Borda count outperforms the Bucklin rule, which in turn outperforms the MedRank rule.

Again because of the slow rate of convergence in Figure 2 we show the graphs only ranging from 3 to 50 voters. The limiting values for the graphs have been determined by Bednay et al. [23]. Figure 2 does not show the graph of the Spearman rank correlation welfare function since it is the same as the graph of the Borda count. Furthermore, the Borda count is the best performing SWF, while the MedRank rule is the worst performing one.

Beside the Spearman footrule and the Spearman rank correlation another prominent distance is the Kendall τ distance, and therefore we also provide the ranking of

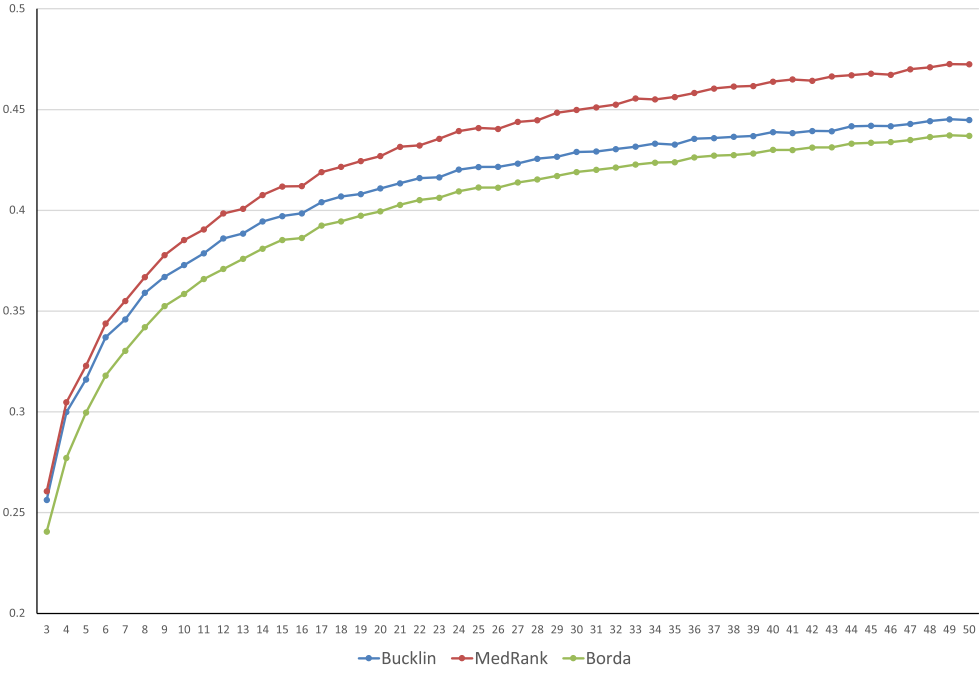


Figure 2. Average distance in case of D_2 and $m = 5$.

the Borda count, the Bucklin rule and the MedRank rule by employing the latter distance. By definition the Kemény–Young SWF minimizes the Kendall τ distance (see [1]). Bartholdi et al. [24] had shown that determining the Kemény–Young SWF ranking is computationally difficult.

Figure 3 shows that based on the Kendall τ distance the closest rule to the Kemény–Young SWF is the Borda count, the second closest is the Bucklin rule, while the one that is the furthest away is the MedRank rule.

The *main message* from this section is that for most of the profiles the ordering of the SWFs is the Borda count, the Bucklin rule and the MedRank rule for all three prominent distances.

6. Spearman’s footrule – MedRank

As we have seen it from the graphs of our simulations in Figure 1 the MedRank algorithm fails to be the Spearman footrule welfare function for most of the profiles. Addressing this question, we recall our result from Bednay et al. [3], which shows that the probability of ties tends to one for the MedRank rule as the number of voters tends to infinity. More precisely, with a probability arbitrarily close to 1 for a sufficiently large number of voters the rank of all alternatives will be the same in case of an odd number of alternatives, while it will be ‘one of the two median’ ranks in case of an even number of alternatives.

Proposition 6.1 (Bednay et al. [3]): *For a given m the probability that the median ranks of the alternatives equals $\lfloor (m+1)/2 \rfloor$ or $\lceil (m+1)/2 \rceil$ tends to 1 if n tends to infinity, where if m is odd then the median is unique and the two specified numbers are the same.*

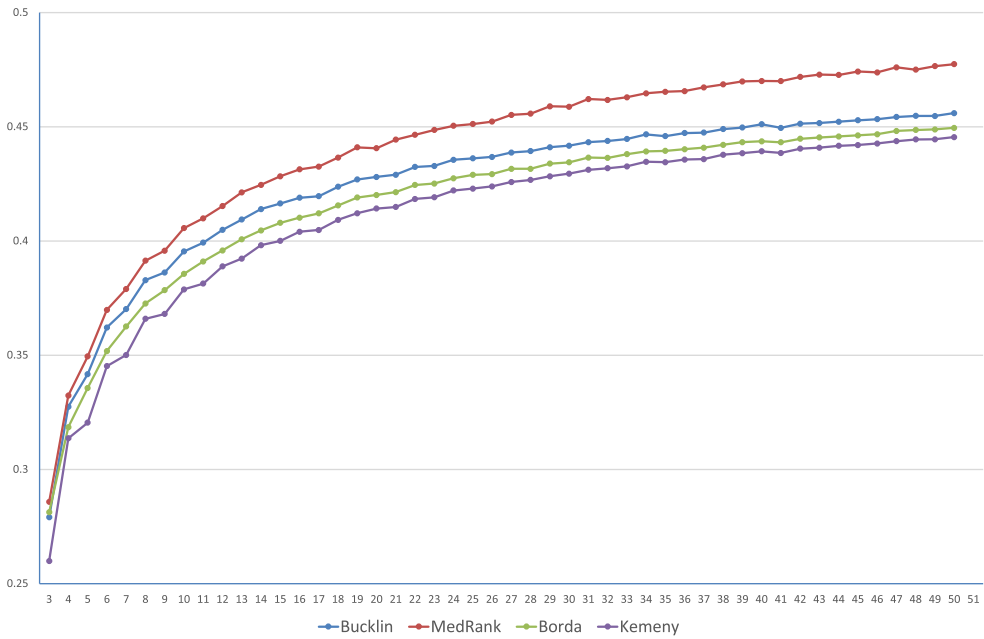


Figure 3. Average distance in case of D_B and $m = 5$.

Table 6. The MedRank rule differs from the Spearman footrule welfare function.

Rank	$\succ_1$	$\succ_2$	$\succ_3$	$\succ_4$	$\succ_5$	$\succ_6$
1	a_1	a_1	a_2	a_2	a_3	a_3
2	a_4	a_4	a_4	a_4	a_4	a_4
3	a_2	a_3	a_1	a_3	a_1	a_2
4	a_5	a_5	a_5	a_5	a_5	a_5
5	a_3	a_2	a_3	a_1	a_2	a_1

Since Proposition 6.1 was published in Hungarian for the sake of completeness we include its proof in English in Appendix 2.

Furthermore, we show that there is no tie-breaking rule that makes the MedRank rule equal to the Spearman footrule welfare function.

Proposition 6.2: *There are profiles for which there does not exist a tie-breaking rule such that the MedRank rule minimizes the Spearman footrule distance.*

Proof: In Table 6 we provide a profile for which we cannot find a tie-breaking rule such that the choice of the Medrank rule minimizes the Spearman footrule distance.

Observe that the median ranks of a_1 , a_2 and a_3 all equal three, while the median ranks of a_4 and a_5 equal 2 and 4, respectively. Therefore, for any tie-breaking rule the MedRank rule would rank a_4 as the highest ranked alternative and a_5 as the lowest ranked alternative.

The related minimum cost matching problem to the profile given in Table 6 is shown in Table 7 from which it can be seen that a_4 and a_5 have to be ranked second and fourth in an arbitrary way, while the alternatives a_1 , a_2 and a_3 have to be ranked first, third and fifth.

Table 7. A possible minimum cost matching gives $a_1 \succ a_4 \succ a_2 \succ a_5 \succ a_3$.

	1	2	3	4	5
a_1	12	10	8	10	12
a_2	12	10	8	10	12
a_3	12	10	8	10	12
a_4	6	0	6	12	18
a_5	18	12	6	0	6

However, such a ranking cannot be obtained by the MedRank rule as explained at the end of the previous paragraph. ■

7. Spearman's rank correlation – Borda count

In this section we show that the Borda count is the Spearman rank correlation minimizing SWF. This strengthens the result of Dwork et al. [2] stating that for any $\succ \in \mathcal{P}^n$ for which the Borda count determines a linear ordering BC minimizes Spearman's rank correlation. Furthermore, it strengthens the result by Bednay et al. [3] showing that the Borda count is the Spearman rank correlation minimizing SWF with probability one in a probabilistic setting.⁴

The goal is to find an ordering $\succ$ which minimizes

$$\begin{aligned}
 d_2(\Pi, \succ) &= \frac{1}{n} \sum_{i=1}^n d_2(\succ_i, \succ) = \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^m (rk[a_j, \succ_i] - rk[a_j, \succ])^2 \\
 &= \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^m (sc[a_j, \succ_i] - sc[a_j, \succ])^2.
 \end{aligned}$$

To simplify our notations, let $r_{ij} = rk[a_j, \succ_i]$ and $x_j = rk[a_j, \succ]$. So the problem is, for a given $(r_{ij})_{(i,j) \in M \times N}$, to find $(x_j)_{j \in M}$ for which

$$\frac{1}{n} \sum_{i=1}^n \sum_{j=1}^m (r_{ij} - x_j)^2$$

is the minimum, where $\mathbf{x}$ is a permutation of $\{1, 2, \dots, m\}$. Note that r_{ij} is also a permutation of $\{1, 2, \dots, m\}$ for a fixed i .

Let's solve the problem more generally: allow for the values r_{ij} any arbitrary real number, and the vector $\mathbf{x}$ is a permutation of arbitrary fixed numbers.

Let $b_j = \sum_{i=1}^n r_{ij}$ be the sum of ranks of alternative j .

Proposition 7.1: *The expression*

$$\frac{1}{n} \sum_{i=1}^n \sum_{j=1}^m (r_{ij} - x_j)^2$$

is minimal (or maximal) if the vectors $\mathbf{x}$ and $\mathbf{b}$ are in the same (or opposite) ordering, i.e. we take the Borda (or inverse Borda) ordering.

Proof: The multiplier $\frac{1}{n}$ can obviously be omitted, so we just consider expression

$$\sum_{i=1}^n \sum_{j=1}^m (r_{ij} - x_j)^2.$$

Carrying out squaring, we get

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^m (r_{ij} - x_j)^2 &= \sum_{i=1}^n \sum_{j=1}^m (r_{ij}^2 - 2r_{ij}x_j + x_j^2) = \sum_{i=1}^n \sum_{j=1}^m r_{ij}^2 - 2 \sum_{j=1}^m \sum_{i=1}^n r_{ij}x_j + n \sum_{j=1}^m x_j^2 \\ &= \sum_{i=1}^n \sum_{j=1}^m r_{ij}^2 - 2 \sum_{j=1}^m b_j x_j + n \sum_{j=1}^m x_j^2. \end{aligned}$$

Note that the first and last parts are constant because the values of r_{ij} are given and we can only change the ordering of x_j . For the term in the middle because of the minus sign, we have to look at the maximum (or minimum) of $\sum_{j=1}^m b_j x_j$. Which, due to the rearrangement inequality, happens when the sequences (b_j) and (x_j) are in the same (or opposite) ordering. ■

Since the definition of the b_j 's before the statement of Proposition 7.1 is not just valid for the Borda count defined in Section 3, but also for any scoring rule, which assigns to any rank a sequence of decreasing non-negative integers according to Baumeister and Rothe [25], Proposition 7.1 is valid for any scoring rule by replacing the Borda scores with the scores given by another scoring method.

8. Conclusion

For a larger number of voters and 3 to 5 alternatives we took a random sample of 2500 profiles. Our main findings were that for many profiles the MedRank algorithm does not minimize the Spearman footrule, while the Borda count minimizes the Spearman rank correlation for all profiles. In addition, we observed that with respect to the Spearman footrule, the Spearman rank correlation and the Kendall τ distances the Borda count performs the best, the Bucklin rule the second best and the MedRank rule the worst. Proposition 6.2 demonstrates that there is no tie-breaking rule that makes the MedRank rule the Spearman footrule welfare function and Proposition 7.1 shows that the Borda count is the Spearman rank correlation welfare function.

Finally, we outline possible further research directions. In our analysis we presented our results for a given number of alternatives m and let the number of voters n vary. Clearly, this is the more interesting case in the social choice context. However, in computer science applications the other case in which we fix n and let m vary can be equally interesting. Since m alternatives imply $m!$ rankings, this analysis is far less tractable. ^{5 6}

Notes

1. The linear ordering selected by an anonymous tie-breaking rule is invariant to the ordering of voters' preferences.

2. The graphs for 3 and 4 alternatives for the Spearman rank correlation and the Kendall τ distance are available upon request from the authors.
3. It is worth mentioning that [23] have determined the limit.
4. Since this result was published in Hungarian for the sake of completeness we include its statement and its proof in English in Appendix 3.
5. This is more admissible than the usual rank assignments since, for instance, there can be two gold medallists in the olympic games, but there cannot be two silver medallists without a gold medallist.
6. We have these cases only if m is odd.

Disclosure statement

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References

- [1] Kemény J. Mathematics without numbers. *Daedalus*. 1959;88:577–591.
- [2] Dwork C, Kumar R, Naor M, et al. Rank aggregation revisited; 2002. [accessed 2022 January 6]. Available from: <http://www.cse.msu.edu/cse960/Papers/games/rank.pdf>
- [3] Bednay D, Fleiner B, Tasnádi A. A MedRank algoritmus korlátai (Limitations of the MedRank Algorithm). *Szigma*. 2024;55:28–33.
- [4] Diaconis P, Graham RL. Spearman's footrule as a measure of disarray. *J R Stat Soc: Series B (Methodological)*. 1977;39:262–268. doi: 10.1111/j.2517-6161.1977.tb01624.x
- [5] Monjardet B. Concordance between two linear orders: the Spearman and Kendall coefficients revisited. *J Classif*. 1997;14:269–295. doi: 10.1007/s003579900013
- [6] Berry KJ, Mielke PW. Spearman's footrule as a measure of agreement. *Psychol Rep*. 1997;80:839–846. doi: 10.2466/pr0.1997.80.3.839
- [7] Farkas D, Nitzan S. The Borda rule and Pareto stability: A comment. *Econometrica*. 1979;47:1305–1306. doi: 10.2307/1911965
- [8] Nitzan S. Some measures of closeness to unanimity and their implications. *Theory Decis*. 1981;13:129–138. doi: 10.1007/BF00134214
- [9] Lehrer E, Nitzan S. Some general results on the metric rationalization for social decision rules. *J Econ Theory*. 1985;37:191–201. doi: 10.1016/0022-0531(85)90036-5
- [10] Campbell DE, Nitzan S.L. Social compromise and social metrics. *Soc Choice Welfare*. 1986;3:1–16. doi: 10.1007/BF00433520
- [11] Elkind E, Faliszewski P, Slinko A. Distance rationalization of voting rules. *Soc Choice Welfare*. 2015;45:345–377. doi: 10.1007/s00355-015-0892-5
- [12] Andjiga NG, Mekuko AY, Moyouwou I. Metric rationalization of social welfare functions. *Math Soc Sci*. 2014;72:14–23. doi: 10.1016/j.mathsocsci.2014.10.001
- [13] Mahajne M, Nitzan S, Volij O. Level r consensus and stable social choice. *Soc Choice Welfare*. 2015;45:805–817. doi: 10.1007/s00355-015-0882-7
- [14] Bednay D, Moskalenko A, Tasnádi A. Does avoiding bad voting rules lead to the good ones? *Operations Research Letters*. 2017;45:448–451. doi: 10.1016/j.orl.2017.07.001
- [15] Bednay D, Moskalenko A, Tasnádi A. Dictatorship versus manipulability. *Math Soc Sci*. 2019;101:72–76. doi: 10.1016/j.mathsocsci.2019.07.001
- [16] Lagerspetz E. Social choice and democratic values. Switzerland: Springer International Publishing; 2016.
- [17] Héberger K. Sum of ranking differences compares methods or models fairly. *Trends Analyt Chem*. 2010;29:101–109. doi: 10.1016/j.trac.2009.09.009
- [18] Sziklai BR, Héberger K. Apportionment and districting by sum of ranking differences. *PLoS ONE*. 2020;15:1–20. doi: 10.1371/journal.pone.0229209
- [19] Lin S. Rank aggregation methods. *WIREs Comput. Stat.*. 2010;2:555–570. doi: 10.1002/wics.v2:5

- [20] Erdélyi G, Fellows MR, Rothe J, et al. Control complexity in Bucklin and fallback voting: A theoretical analysis. *J Comput Syst Sci.* 2015;81:632–660. doi: [10.1016/j.jcss.2014.11.002](https://doi.org/10.1016/j.jcss.2014.11.002)
- [21] Puckelsheim F. Proportional representation: apportionment methods and their application. Switzerland: Springer International Publishing; 2014.
- [22] Lee PH, Yu PL. An R package for analyzing and modeling ranking data. *BMC Med Res Methodol.* 2013;13:65. doi: [10.1186/1471-2288-13-65](https://doi.org/10.1186/1471-2288-13-65)
- [23] Bednay D, Fleiner B, Tasnádi A. An indifference result for social choice rules in large societies. *Eur J Oper Res.* 2025;321:208–213. doi: [10.1016/j.ejor.2024.09.018](https://doi.org/10.1016/j.ejor.2024.09.018)
- [24] Bartholdi J, Tovey CA, Trick MA. Voting schemes for which it can be difficult to tell who won the election. *Soc Choice Welfare.* 1989;6:157–165. doi: [10.1007/BF00303169](https://doi.org/10.1007/BF00303169)
- [25] Baumeister D, Rothe J. Preference aggregation by voting. In: Rothe, J. (ed.), *Economics of computation*. Berlin: Springer, 197–325. 2016.
- [26] Arratia R, Gordon L. Tutorial on large deviations for the binomial distribution. *Bull. Math. Biol.* 1989;51:125–131. doi: [10.1016/S0092-8240\(89\)80052-7](https://doi.org/10.1016/S0092-8240(89)80052-7)
- [27] Billingsley P. Probability and measure. New York: Wiley; 1995.

Appendices

Appendix 1

Figures A1 and A2 contain the same figures for $m = 3$ and $m = 4$ as Figure 1 in Section 5 for $m = 5$. In all four figures we can observe that the Borda count comes closest to the Spearman footrule welfare function, while surprisingly the MedRank algorithm is the worst performing one.

Appendix 2

Proof of Proposition 6.1.: We introduce the MedRank* algorithm, henceforth briefly MR^* , algorithm which assigns to each alternative a its median rank h_a as defined in Section 4. More

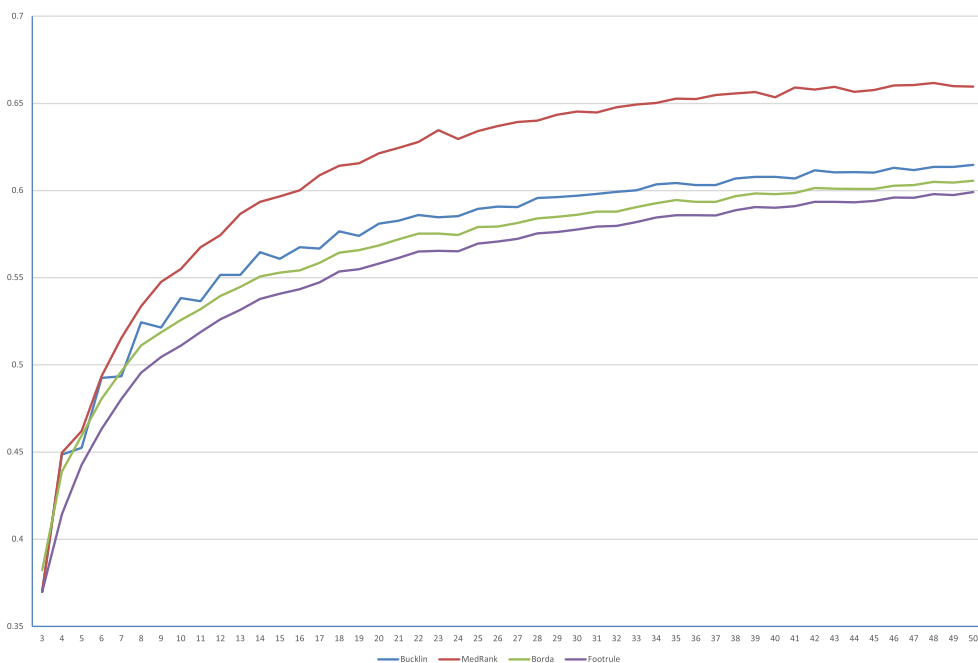


Figure A1. Average distance in case of D_1 and $m = 3$.

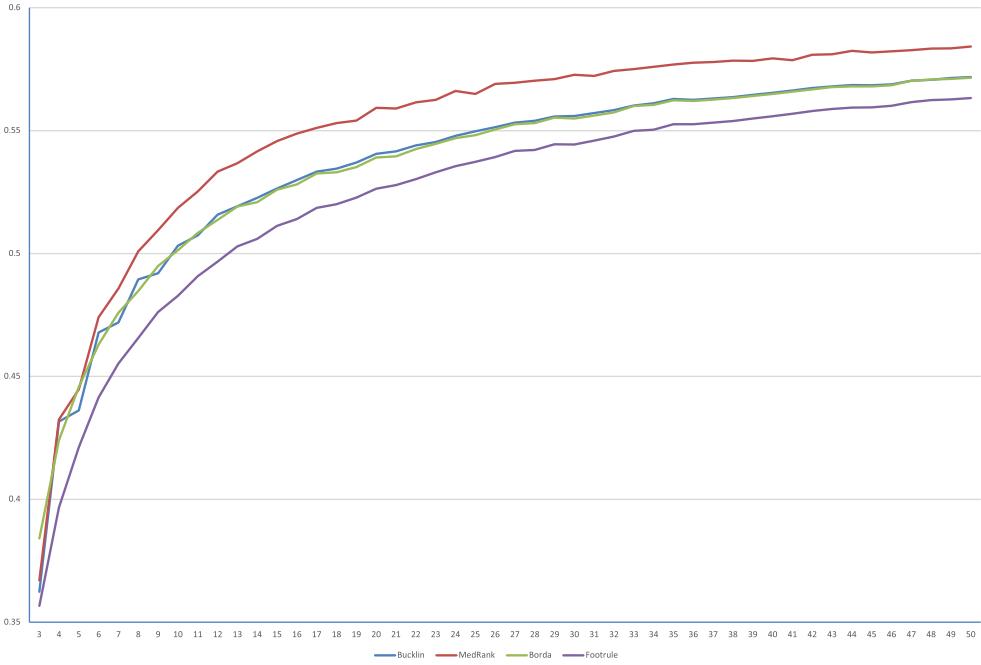


Figure A2. Average distance in case of D_1 and $m = 4$.

specifically, MR^* selects in case of an even number of alternatives the latter rank from the two in the middle. Hence, MR^* maps from $\mathcal{P}^n$ to $\{1, \dots, m\}^A$ and is not an SWF since it can result in weak preferences. In addition, it assigns ranks to the alternatives and not just a weak ordering of them.⁵ Put another way, the MR^* algorithm assigns the rankings of the MR algorithm before resolving potential ties between alternatives. Nevertheless, $D_1(MR^*)$ can be still defined in the same way as for SWFs.

Furthermore, since for any $j = 1, \dots, m$ and any $\Pi \in \mathcal{P}^n$ the solution of

$$\min_{r=1, \dots, m} \sum_{i \in N} |r - rk[a_j, \succ_i]| \quad (A1)$$

is the median of the ranks $rk[a_j, \succ_1], \dots, rk[a_j, \succ_n]$ by the standard property of the minimum of the sum of absolute differences from a set of given numbers. Note that for any j and any Π we have independent minimization problems since for given j and profile Π the minimum point of (A1) equals $rk[a_j, MR^*(\Pi)]$, and therefore for any anonymous SWF F we get

$$\begin{aligned} \sum_{\Pi \in \mathcal{P}^n} \sum_{j=1}^m \min_{r=1, \dots, m} \sum_{i \in N} |r - rk[a_j, \succ_i]| &= \sum_{\Pi \in \mathcal{P}^n} \sum_{j=1}^m \sum_{i \in N} |rk[a_j, MR^*(\Pi)] - rk[a_j, \succ_i]| \\ &\leq \sum_{\Pi \in \mathcal{P}^n} \sum_{j=1}^m \sum_{i \in N} |rk[a_j, F(\Pi)] - rk[a_j, \succ_i]|, \end{aligned}$$

after changing the ordering of the summations and multiplications by the respective constants.

Our next step is to determine for notational convenience the asymptotic distribution of $MR^*(\Pi)$ as n tends to infinity, which is the same as the asymptotic distribution of $MR^*(\succ_{-i})$. Pick an arbitrary alternative a and note that for each $\succ_l$, where $l = 1, \dots, n$, alternative a is ranked in the top r positions with probability r/m . If for a given n we denote by $X_n^{(r)}$ the number of voters ranking a in the top r positions, then $X_n^{(r)}$ is binomially distributed with parameters n and r/m . Hence, the

probability that a is ranked in the top r positions by MR^* equals

$$P(X_n^{(r)} \geq \lfloor n/2 \rfloor + 1) = \sum_{i=\lfloor n/2 \rfloor + 1}^n \binom{n}{i} \left(\frac{r}{m}\right)^i \left(1 - \frac{r}{m}\right)^{n-i} \quad (\text{A2})$$

and the probability that a is ranked r th by MR^* equals

$$P(MR^* = r) = P(X_n^{(r)} \geq \lfloor n/2 \rfloor + 1) - P(X_n^{(r-1)} \geq \lfloor n/2 \rfloor + 1).$$

We recall the result on the binomial distribution in Arratia and Gordon [26], Theorem 1), which we will employ to give an upper bound of the probability in (A2). Let Y_n be binomially distributed with parameters n and p . Then for any $n = 1, 2, \dots$ and any $p < \alpha < 1$ we have

$$P(Y_n \geq \alpha n) \leq e^{-n\left(\alpha \log \frac{\alpha}{p} + (1-\alpha) \log \frac{1-\alpha}{1-p}\right)}. \quad (\text{A3})$$

In our case p takes values $1/m, 2/m, \dots, (m-1)/m$ and because of symmetry we only need to give an upper bound of the probabilities in (A2) for values $1/m, 2/m, \dots, \lfloor m/2 \rfloor /m$ since we can give related required lower bounds on the probabilities in (A2) for values $(\lfloor m/2 \rfloor + 1)/m, \dots, (m-1)/m$ by considering the complementary event. For the missing case $p = 1/2$ if m is even, we will determine (A2) directly. Therefore, we start with the case of $p < \alpha = 1/2$. Substituting these values into (A3), we get

$$P(Y_n \geq \frac{1}{2}n) \leq e^{-\frac{1}{2}n\left(\log \frac{1}{2p} + \log \frac{1}{2(1-p)}\right)} = e^{-\frac{1}{2}n \log \frac{1}{4p(1-p)}},$$

which in turn implies by $p(1-p) < 1/4$ that $P(Y_n \geq \frac{1}{2}n)$ tends to zero as n tends to infinity. Therefore, we also have that for $r = 1, 2, \dots, \lfloor (m-1)/2 \rfloor$ in case of m is even and for $r = 1, 2, \dots, \lfloor m/2 \rfloor$ in case of m is odd, the probabilities in (A2) tend to zero as n tends to infinity. Now if m is even, that is $m = 2k$, then

$$\sum_{i=\lfloor n/2 \rfloor + 1}^n \binom{n}{i} \left(\frac{1}{2}\right)^i \left(1 - \frac{1}{2}\right)^{n-i} = \frac{1}{2^{n+1}} 2^n = \frac{1}{2}.$$

From our limiting results and symmetry (by considering the complementary events) we conclude that when n tends to infinity that the probabilities in (A2) tend to $1/2$ and 1 for $r = m/2$ and $r = m/2 + 1$,⁶ respectively, while for r smaller than the median(s) and for r larger than the median(s) they tend to zero and one as n tends to infinity, respectively. ■

Appendix 3

The following proposition basically implies that the Borda count can differ from the Spearman rank correlation choice just for a smaller and smaller proportion of profiles.

Proposition A.1 (Bednay et al. [3]): *The probability that an additional voter changes the ranking obtained by the Borda count tends to zero as n tends to infinity.*

Proof: For notational convenience we let $i = n$ be the additional voter, which can be assumed without loss of generality by the anonymity of BC. We will show that the ratio of profiles on which $BC(\succ_{-n})$ does not equal $BC(\Pi)$ tends to zero as n tends to infinity. In order to show that we start with picking two arbitrary distinct alternatives a_j and a_k . Let us consider the random walk defined by the difference in the sum of ranks of alternatives a_j and a_k . In particular, we consider the random variable $X_n = \sum_{i=1}^{n-1} (rk[a_j, \succ_i] - rk[a_k, \succ_i])$, where $n = 0, 1, \dots$. Note that $-m < X_n < m$ is a necessary condition for $a_j BC(\succ_{-n}) a_k$ and $a_k BC(\Pi) a_j$. The state space of the random walk, which is a Markov process, is $S = \mathbb{Z}$, the initial state is $X_0 = 0$, the transition probabilities are $p_{s,s+d} = P(X_{i+1} = s+d \mid X_i = s) = (m-d)/(m(m-1))$ if $d = 1, \dots, m-1$ and $p_{s,s-d} = P(X_{i+1} = s-d \mid X_i = s) = (m-d)/(m(m-1))$ if $d = 1, \dots, m-1$ for all $i = 0, 1, \dots, n-1$. Let us denote by $p_{ij}^{(t)}$ the transition probability from state i to state j in t steps. The defined random walk is

1. persistent, meaning that it will return to the starting state sometime with probability one, which can be verified by Billingsley [27] Theorem 8.2.(ii) since $\sum_{t=1}^{\infty} p_{ij}^{(t)} > \frac{1}{m} \sum_{t=1}^{\infty} \tilde{p}_{ij}^{(t)} = \infty$, where $\tilde{p}_{ij}^{(t)}$ stands for the respective transition probabilities of the symmetric Bernoulli random walk,
2. irreducible, that is all states can be reached from every other state in a finite number of transitions and
3. aperiodic since with the exception of X_1 we have $P(X_i = 0) > 0$ for all $i \geq 2$.

By Billingsley [27] Theorem 8.8. we know that $\lim_{n \rightarrow \infty} p_{ij} = 0$ for all i and j . Therefore, for any arbitrarily small $\varepsilon > 0$ and for sufficiently large n we have $\varepsilon/(m(m-1)) > P(-m < X_n < m)$. Now even considering all distinct pairs of alternatives the probability that adding an n th voter changes the result of BC is less than ε and tends to zero as n tends to infinity. ■

We formulate a simple corollary of Proposition A.1.

Corollary A.1 (Bednay et al. [3]): *The probability that BC has tied alternatives or put it otherwise alternatives with identical rank sums tend to zero as n tends to infinity.*

Dwork et al. [2] have shown that for any $\Pi \in \mathcal{P}^n$ for which the Borda count determines a linear ordering BC minimizes Spearman's rank correlation. Taking their result and Corollary A.1 into consideration, we know that for any anonymous SWF F we must have

$$\lim_{n \rightarrow \infty} D_2(BC_\tau) \leq \lim_{n \rightarrow \infty} D_2(F).$$